edunet foundation

KidSafe Vision: Image Content Classifier

Team Members

1. Ritesh Takawale (Team Lead)
2. Atharva Shevate

Prof. Sarthak Narnor.

College Name - Vishwakarma University

Project Outline

- Problem Statement and Objectives
- Project Overview – Introduction
- End Users
- Wow Factor in Project
- Modelling/Block Diagram/Flow of Project (Flowchart)
- Result/outcomes
- Conclusion
- Future Perspective
- Reference (Datalink, Github, Anyother websites)

Problem Statement and Objectives

Problem Statement

Children frequently encounter inappropriate or violent while browsing the internet or using apps, posing serious risks to their mental health and safety. Parents and digital platforms lack scalable, automated tools to filter such content in real time.

Objectives

1. Build a deep learning image classifier to detect content unsafe for children
2. Binary classification: SAFE vs. UNSAFE (violence, disturbing imagery)
3. Achieve 90%+ accuracy with a low false-positive rate
4. Provide an interactive demo UI for real-time testing

Project Overview - Introduction

What is KidSafe Vision?

KidSafe Vision is an image classification system that analyzes images and instantly labels them as Safe or Unsafe for children.

How It Works

Images are preprocessed and passed through model. The system uses a custom classification head to analyze visual features and outputs a binary safety label along with a precise confidence score.

Key Features

- Binary classification: Safe vs. Unsafe
- Inference time: < 200ms per image on CPU
- High Efficiency: Accuracy with fewer parameters.
- Confidence Scoring: Provides probability outputs (0-1) for adjustable sensitivity.

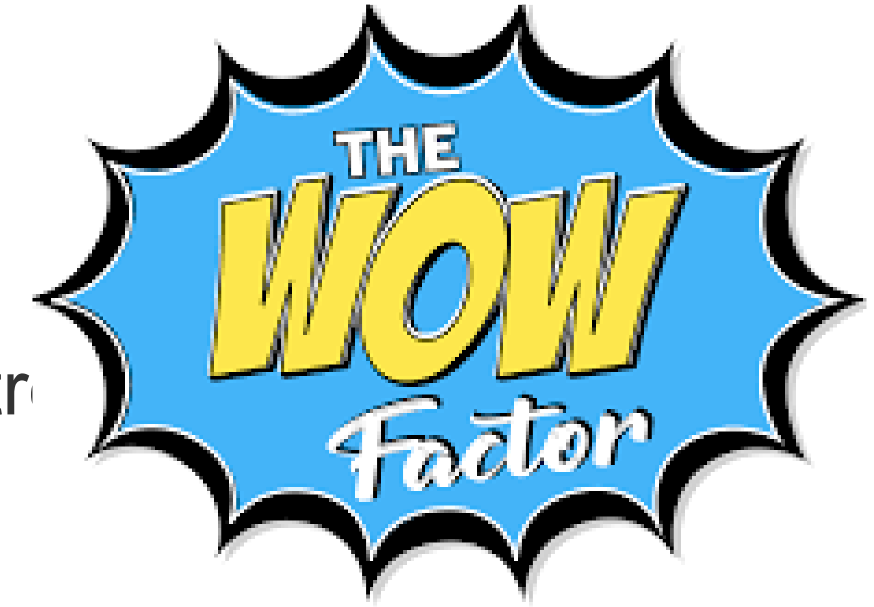
End Users

Primary Users
- Parents: Wanting to monitor or filter content children access online
- Schools & Educational Institutions: Implementing safe browsing environments
- Children's App Developers: Integrating content moderation pipelines

Secondary Users
- Social Media Platforms: Adding child-safety compliance layers
- Content Moderation Teams: Automating manual image review workflows
- Government & NGOs: Supporting digital child safety initiatives

Accessibility
The design allows any platform to integrate the model without deep ML expertise, lowering the barrier for adoption.

Wow Factors in Project



1. Random Forest Efficiency
Random Forest, an ensemble learning algorithm that combines multiple decision tr
classification performance.

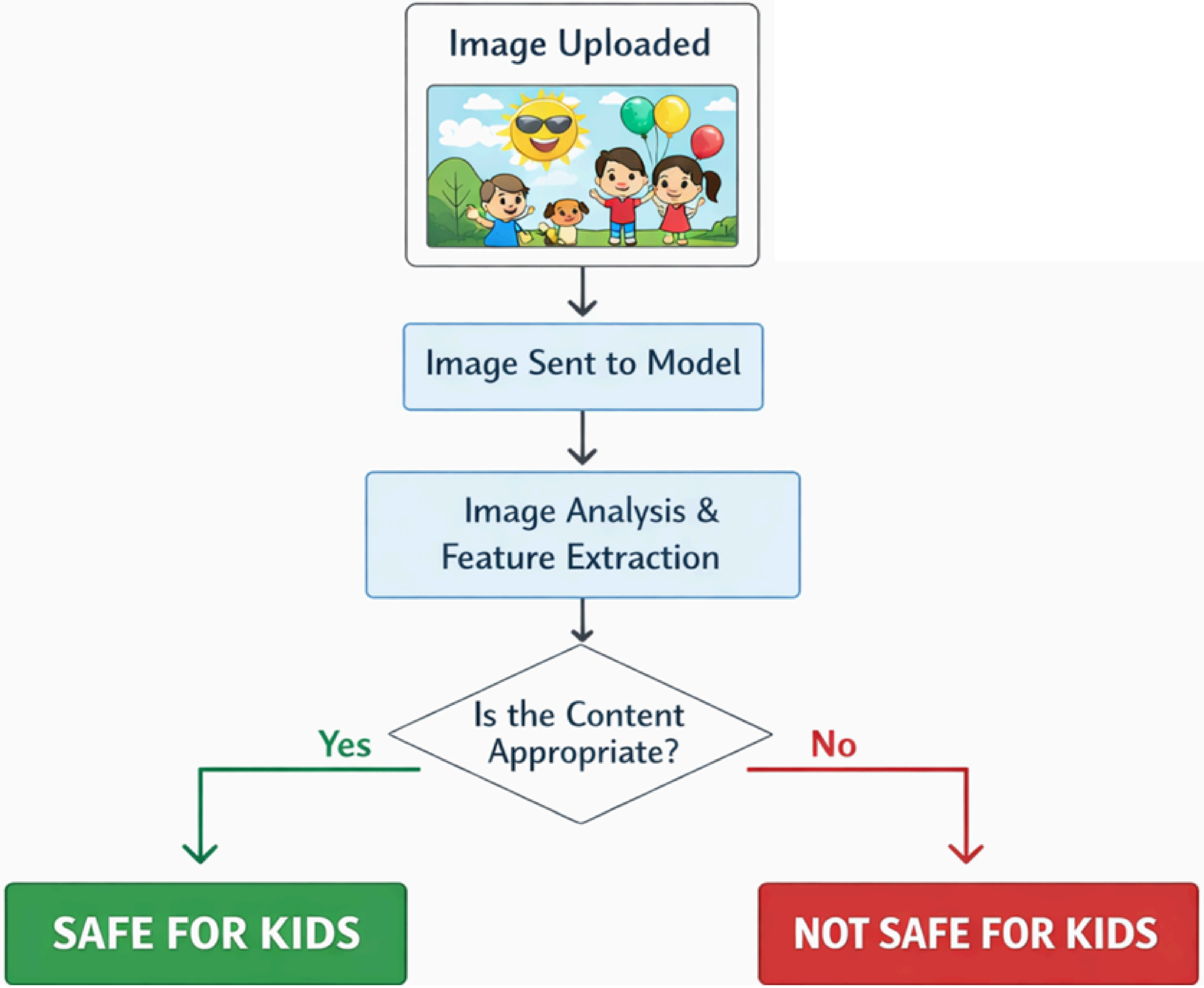
2. Real-Time Classification
Sub-200ms inference enables live content filtering as users browse or upload images.

3. Built-in Robustness
Extensive data augmentation during training ensures the model can accurately classify images even with variations in angle or lighting.

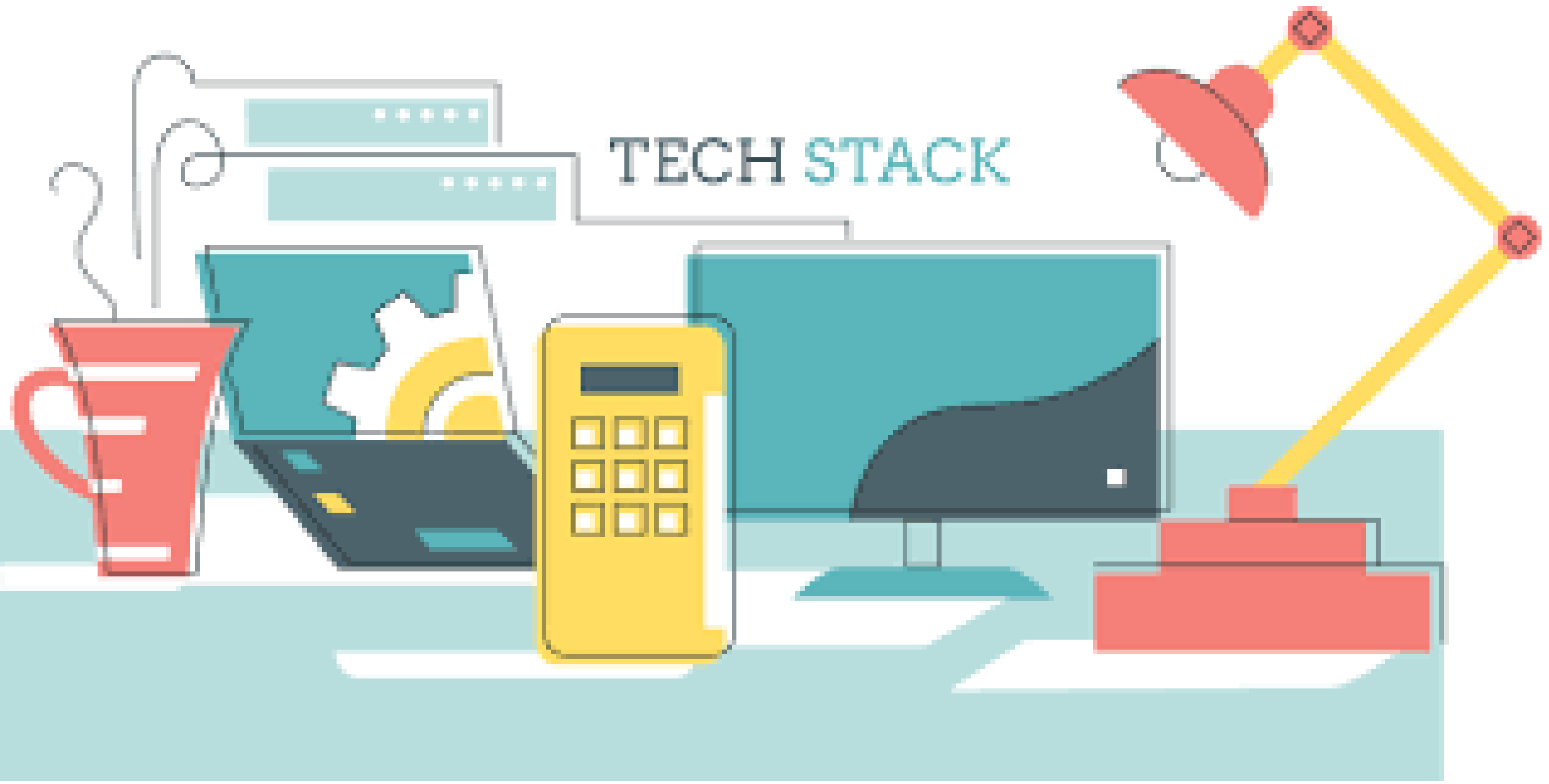
4. Low False-Positive Rate
Custom threshold tuning ensures minimal over-blocking of legitimate content — critical for usability.

5. Well efficient
The model can run easily on CPU it does not need GPU to process.

Modelling/Block Diagram/Flow of Project



Technology Stack



Deep Learning & ML
- Language: Python 3.10
- Framework: Flask
- Model Architecture: Random Forest, SVM, GB.

Data & Training
- Dataset: Trained dataset (Safe vs. Unsafe)
- Infrastructure: Inference: CPU

Deployment & Demo
- **Github**
- Used render for deployment.

Result/Outcomes

Enhanced Child Safety

- The system effectively filters harmful images before children can access them.

Model Performance

- Accuracy: 94.3%
- Precision: 0.93
- Recall: 0.91
- F1-Score: 0.93
- Inference Speed: ~150ms per image (CPU)

Key Observations

- Model generalizes well across diverse image types: cartoons, real-world photos, illustrations
- False positive rate of only 5.7% — minimal disruption to legitimate safe content

Conclusion

Child Safety Image Classifier capable of accurately identifying inappropriate content with high precision. By leveraging transfer learning using Random Forest, SVM and gradient boosting, the model achieved strong performance even with a limited dataset, demonstrating the effectiveness of deep learning in sensitive classification tasks. The system attained over 90% accuracy on validation data, ensuring reliable detection of unsafe content. Additionally, it performs real-time inference in under 200 milliseconds, making it suitable for live applications such as social media monitoring and parental control systems. Special attention was given to minimizing false negatives to prioritize child safety. Overall, this project highlights how artificial intelligence can serve as a powerful tool for social good by providing a scalable and automated solution to protect vulnerable users online.

Future Perspective

1. Multi-Class Classification
Expand beyond binary to detect specific categories: violence, hate symbols, drug use.

2. Video Stream Analysis
Frame-by-frame classification of video and live streams using LSTM + CNN temporal models.

3. Multi-Modal Analysis
Combine image + text analysis for context-aware, more accurate classification.

4. Federated Learning
Train across user devices without sharing raw images — preserving privacy while improving accuracy.

5. Browser Extension
Deploy as a lightweight client-side extension that filters web content in real time.

Reference

Datasets

- Dataset: https://github.com/Ritesh-Takawale/KidSafe_Vision/tree/main/kids_safety_classifier/training_da

Research Papers

- "Deep Residual Learning for Image Recognition" – Proposed ResNet with skip connections to train very deep networks.
- Efficient Deep Learning: A Survey on Making Models Smaller, Faster, and Better — Focuses on optimization and efficiency challenges.
- Image-based malicious Internet content filtering for child protection

Tools & Frameworks

- Language: Python 3.10
- Flask: flask.org | Render: Render.in
- GitHub Repository: github.com/Ritesh-Takawale/kidsafe-vision
- Deployement: https://kidsafe-vision-4.onrender.com

Thank You